

Questions — Organizational Learning from Innovation

Analytical Questions

1. Distinguish between parameter updating and model restructuring as forms of organizational learning. Provide an example of each from a real company, and explain how the organization should determine which type of learning is needed.
2. Why does behavioral evidence (what customers do) typically provide higher-precision feedback than stated evidence (what customers say)? Use the Active Inference framework to explain the gap between self-models and behavior, and discuss how this gap affects feedback loop design.
3. Analyze the Lean Startup's "Build-Measure-Learn" cycle as an Active Inference perception-action loop. Map each step to the corresponding Active Inference component, and explain what "validated learning" means in terms of model evidence.
4. How does the SECI model (Socialization, Externalization, Combination, Internalization) describe the conversion between tacit and explicit organizational knowledge? What are the challenges of each conversion step, and how does each relate to Active Inference model parameters?
5. Explain why premature pivoting (restructuring the model when parameter updating would suffice) is as dangerous as delayed pivoting (updating parameters when the model structure is wrong). How does Bayesian model comparison help determine the correct response?
6. Analyze Peter Senge's five disciplines of the learning organization through the Active Inference framework. How does each discipline contribute to the organization's capacity for model generation, testing, and updating?

7. How does Google's "20% time" policy balance exploitation and exploration at the organizational level? What are the Active Inference implications of dedicating organizational resources to model exploration rather than model optimization?
8. Explain why post-mortems (retrospectives) are valuable from a knowledge management perspective. What information must a post-mortem capture to serve as effective organizational long-term memory?
9. How does Amazon's "two-pizza team" structure create a distributed Active Inference architecture? What are the advantages and disadvantages of distributing model inference across many small teams versus centralizing it in a large planning group?
10. Analyze the concept of "innovation accounting" — measuring validated learning rather than traditional metrics. Why might an organization that is learning rapidly (reducing prediction errors) still appear to be failing by traditional metrics? How should investors and stakeholders interpret innovation accounting data?

Applied Questions

1. Design a prediction error inventory for your invention's first six months post-launch. List your five most important predictions, assess their confidence levels, and specify how you would detect each type of error.
2. Design a customer feedback loop for your invention that distinguishes between high-precision and low-precision evidence. Specify at least five feedback sources, rank them by precision, and explain how you would weight each source in your model updating process.
3. Construct a realistic pivot scenario for your invention. Describe the evidence that would accumulate against your current model, propose an alternative model, and conduct a formal Bayesian model comparison to determine whether the pivot is justified.
4. Design a knowledge management system for your inventive organization. Address how you would capture both tacit and explicit knowledge, how you would ensure knowledge persists across team changes, and how you would make stored knowledge accessible and actionable.

5. Assess your inventive organization against Senge's five learning organization disciplines. For each discipline, rate your current capability and design one specific improvement action.
6. Your invention has been on the market for three months and customer retention is 20% lower than predicted. Design a diagnostic investigation: what data would you collect, what hypotheses would you test, and what threshold would trigger a pivot versus a parameter adjustment?
7. Compare two approaches to organizational learning: Spotify's squad model (autonomous teams with independent models) and a centralized product development approach (single model maintained by leadership). What are the conditions under which each approach produces faster learning?
8. Design an A/B testing strategy for your invention that tests your three most uncertain model predictions. For each test, specify the hypothesis, the control and treatment conditions, the success metric, the sample size needed, and the decision you would make based on each possible outcome.
9. Your key engineer (who holds significant tacit knowledge about a critical technology) is leaving the organization. Design a knowledge transfer process that captures as much of their expertise as possible. How do you handle the inherently tacit components that resist explicit documentation?
10. Create an "organizational learning dashboard" with five metrics that track your organization's learning rate. For each metric, explain what it measures, how it would be collected, and what actions would be triggered by changes in each metric.